

Computing Koopman Modes

(1)

Consider the discrete-time dynamical system:

$$X_{k+1} = F(X_k), \quad k=0,1,2,\dots,$$

and the associated Koopman operator $[K_g] = [g \circ F]$ that governs the dynamics of observables $g: \mathbb{R}^d \rightarrow \mathbb{C}$ via

$$g_{k+1} = K g_k, \quad k=0,1,2,\dots$$

$\Rightarrow$ Eigenpairs of K encode information about F .

~~$\Rightarrow$ If K has $\lambda_1, \lambda_2, \dots, \lambda_n$~~

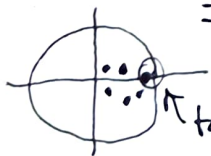
Given observable $g(x) = c_1 v_1(x) + \dots + c_n v_n(x)$, where

$$K v_j = \lambda_j v_j, \quad j=1,\dots,n,$$

Can we extract information about ~~g~~ eigenvectors ^{from} ~~(g, K)~~

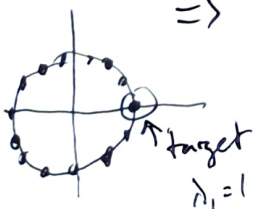
$$K_m = \{g, K g, \dots, K^{m-1} g\}?$$

$\Rightarrow$ If $|\lambda_1| > |\lambda_2| > \dots > |\lambda_n|$, can use "power method":



$$K^{m-1} g \approx \lambda_1^{m-1} c_1 v_1 \quad \Rightarrow \quad w_m = \frac{K^{m-1} g}{\|K^{m-1} g\|} \xrightarrow{m \rightarrow \infty} v_1$$

$\Rightarrow$ If $|\lambda_1| = |\lambda_2| = \dots = |\lambda_n| = 1$ and $\lambda_1 = 1$, use "Fourier" Average:



$$w_m = \frac{1}{m} [g + K g + \dots + K^{m-1} g] \xrightarrow{m \rightarrow \infty} v_1$$

(2)

Fourier/Laplace/Birkhoff Averages

Let's analyze the behavior of w_m as $m \rightarrow \infty$.

$$w_m = \frac{1}{m} [1 + \lambda_1 + \dots + \lambda_1^{m-1}] c_1 v_1 \\ + \frac{1}{m} [1 + \lambda_2 + \dots + \lambda_2^{m-1}] c_2 v_2 \\ \vdots \\ + \frac{1}{m} [1 + \lambda_n + \dots + \lambda_n^{m-1}] c_n v_n$$

$\Rightarrow$ How does $f_m(\lambda) = \frac{1}{m} \sum_{j=0}^{m-1} \lambda^j$ behave as $m \rightarrow \infty$ on each $\{\lambda_j\}_{j=1}^n$?

$\lambda_1 = 1$ | If $\lambda_1 = 1$, $f_m(\lambda_1) = \frac{1}{m} \sum_{j=0}^{m-1} 1^j = \frac{m}{m} = 1$

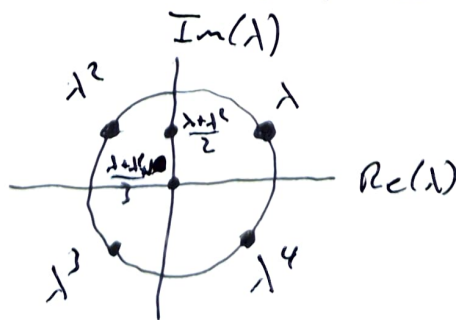
$|\lambda_j| = 1$ | If $\lambda \neq 1$ but $|\lambda| = 1$, then we calculate directly

$$f_m(\lambda) = \frac{1}{m} (1 + \lambda + \dots + \lambda^{m-1}) = \frac{1}{m} \frac{1 - \lambda^m}{1 - \lambda} \quad (\lambda \neq 1)$$

Since $|\lambda| = 1$, we have $|1 - \lambda^m| \leq 1 + |\lambda|^m \leq 2$, so

$$|f_m(\lambda)| \leq \frac{1}{m} \frac{2}{|1 - \lambda|} \rightarrow 0 \text{ as } m \rightarrow \infty.$$

Intuitively, λ^m cycles around unit circle, as $m = 0, 1, 2, \dots$, so their average tends to move closer to zero the origin.



Targeting other eigenvalues

Targeting $\lambda_1 = 1$ is useful for computing ^(or) invariants and conserved quantities of the dynamical system, such as the energy observable for the nonlinear pendulum. For other applications, like isochrons of a stable limit cycle w/ natural frequency ω_0 , we may want other eigenvalues on the unit circle, $\lambda_{\omega_0} = \exp(i k \omega_0 \Delta t)$.

Q: Can we adapt the time averaging strategy to compute λ_{ω_0} ?

$\Rightarrow$ Suppose $\lambda_1 = \lambda_{\omega_0}$ with v_1 is the target, and set the phase $\theta = k \omega_0 \Delta t$, so $\lambda_1 = e^{i\theta}$.

$\Rightarrow$ ~~Modify~~ Set $g_j = K^{j\theta} g_e \rightarrow e^{-i\theta j} g_j = e^{-i\theta j} K^{j\theta} g_e$

$\Rightarrow$ Perform averaging as before:

$$\omega_m = \frac{1}{m} \left[g + e^{-i\theta} K g + \dots + e^{-i(m-1)\theta} K^{m-1} g \right]$$

$$= \frac{1}{m} \left[1 + e^{-i\theta} \lambda_1 + \dots + (e^{-i\theta} \lambda_1)^{m-1} \right] c_1 v_1$$

$$+ \frac{1}{m} \left[1 + e^{-i\theta} \lambda_2 + \dots + (e^{-i\theta} \lambda_2)^{m-1} \right] c_2 v_2$$

$$\vdots$$

$$+ \frac{1}{m} \left[1 + e^{-i\theta} \lambda_n + \dots + (e^{-i\theta} \lambda_n)^{m-1} \right] c_n v_n$$

$$|\lambda| = 1 \quad \Rightarrow \quad \sum_m (e^{-i\theta} \lambda)^j = \begin{cases} 1 & \lambda = e^{i\theta} \\ \frac{1}{m} \frac{1 - (e^{-i\theta} \lambda)^m}{1 - e^{-i\theta} \lambda} & \lambda \neq e^{i\theta} \end{cases}$$

$$\lim_{m \rightarrow \infty} \left| \sum_m (e^{-i\theta} \lambda)^j \right| = \begin{cases} 1 & \lambda = e^{i\theta} \\ 0 & \lambda \neq e^{i\theta} \end{cases}$$

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Convergence Rates/Error Estimates

To estimate the convergence rate, we write ~~the matrix~~

$$W_m - c_1 f_m(\lambda_1) V_1 = c_2 f_m(\lambda_2) V_2 + \dots + c_n f_m(\lambda_n) V_n$$

triangle inequality $\Rightarrow \|W_m - c_1 f_m(\lambda_1) V_1\| \leq c_2 |f_m(\lambda_2)| + \dots + c_n |f_m(\lambda_n)|$

$$\leq \max_{2 \leq j \leq n} c_j |f_m(\lambda_j)|$$

$$\leq \frac{1}{m} \frac{\|g\|_\infty}{\|1\|_\infty} \frac{2 \max_{2 \leq j \leq n} |c_j|}{\min_{2 \leq j \leq n} |1 - \lambda_j|}$$

"distribution" of g over eigenspaces

$\uparrow$ "slow decay w/ $m \rightarrow \infty$ "

$\Rightarrow$ If $\{V_j\}_i$ are far from orthogonal, $\{c_j\}_i$ can be very large even if $\|g\|$ is not. This is a common feature for non-normal matrices / operators, which makes eigenvalue analysis of algorithms like power method and averaging misleading.

$\Rightarrow$ Can derive other polynomial filters to target ~~reg~~ eigenvalues in particular regions.

$\Rightarrow$ Krylov methods have drawbacks in this setting

- ~~non~~ a posteriori orthogonalization
- non-normality
- instability